



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Mohit

29/06/2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



Executive Summary

Objective:

Predict whether a SpaceX Falcon 9 first stage landing will be successful.

Approach

The project combined data collection, data wrangling, exploratory data analysis, interactive visual analytics, and machine learning classification models.

Key Results

- Launch success depends on payload, orbit type, and launch site.
- Interactive dashboards provided operational insights.
- Decision Tree achieved the highest validation accuracy.



Introduction

Background

Space X advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch.

Problem

Can historical launch data be used to predict whether a Falcon 9 booster will successfully land? If landing success can be predicted accurately, launch costs and mission planning can be optimized.

Project Goal

Develop data-driven model capable of predicting Falcon 9 landing success.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Collected SpaceX Falcon 9 launch data using the SpaceX API and web scraping.
- Perform data wrangling
 - Cleaned missing values and selected the relevant features.
 - Created a binary encoded landing outcome.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

The dataset was collected from mainly two data sources and multiple techniques:

- Launch data was collected using **GET requests** to the **SpaceX API** .
- The API response was then decoded in **JSON** format using the `.json()` method and converted into a **Pandas DataFrame** using `pandas.json_normalize()`.
- The dataset was cleaned by handling missing values that is replacing the missing values with mean or average value of the column.
- Additional Falcon 9 launch information was collected through **web scraping** from the **Wikipedia Falcon 9 launch records** using the **BeautifulSoup** library.
- The HTML tables were parsed, extracted into a structured format, and converted into a **Pandas DataFrame**.

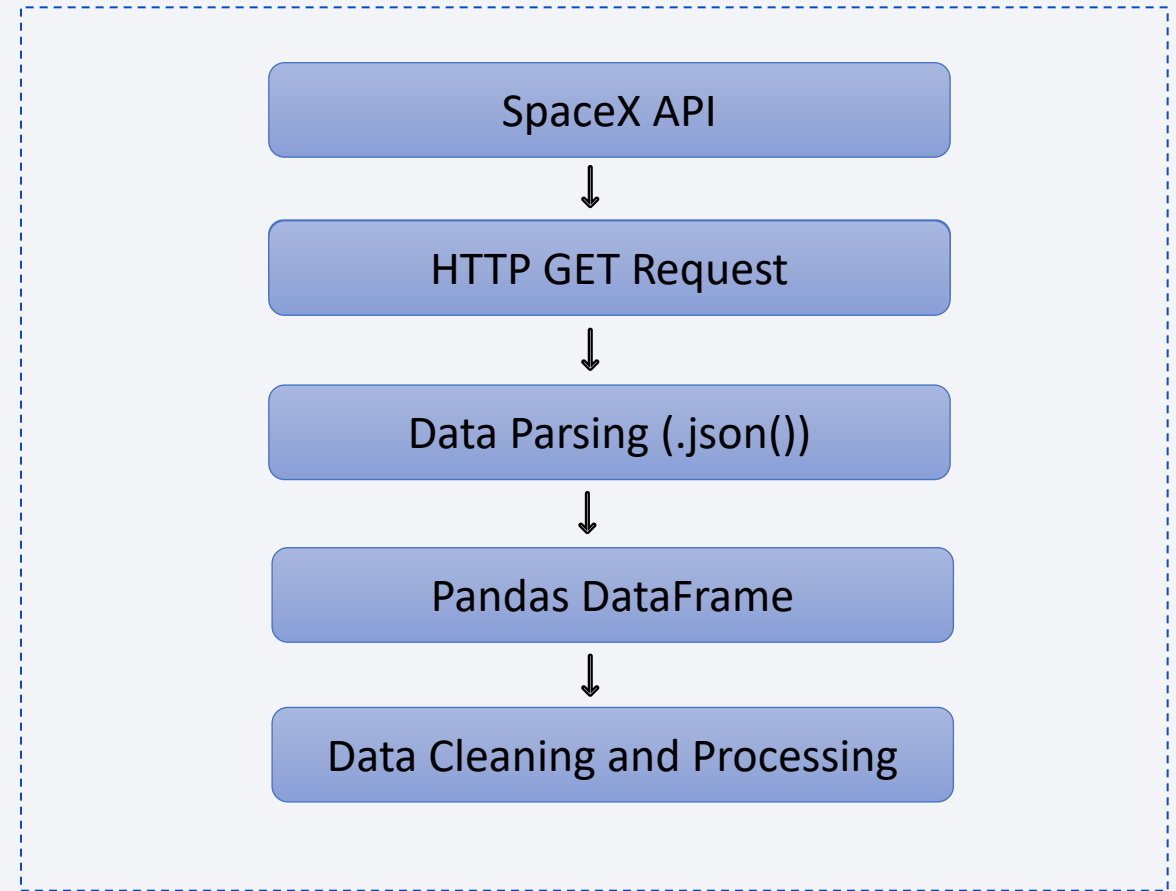
Data Collection- SpaceX API



- Launch data was collected from the SpaceX API by sending HTTP GET requests. The JSON responses were parsed and converted into a structured Pandas DataFrame, providing launch details for further analysis.

GitHub URL of the notebook:

[https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon 9 Data Collection API.ipynb](https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon%209%20Data%20Collection_API.ipynb)

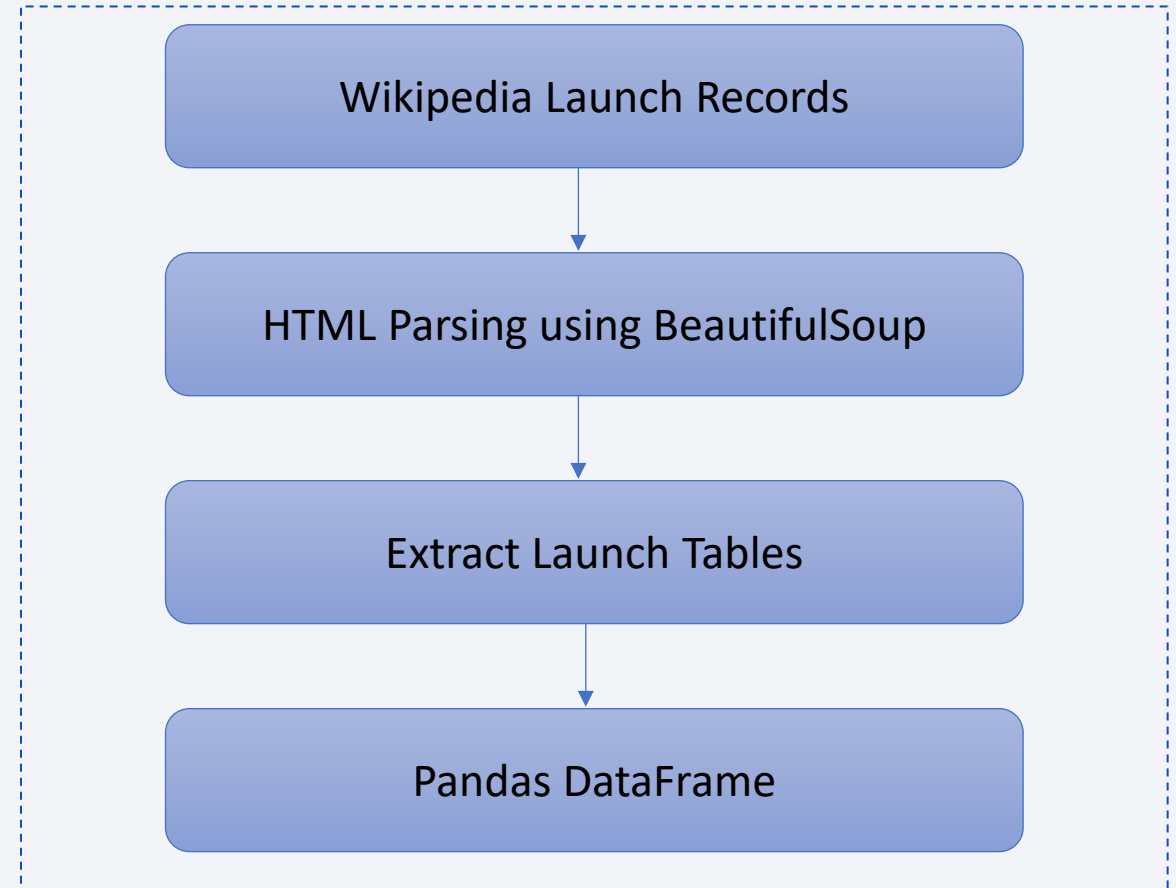


Data Collection - Scraping

- Additional Falcon 9 launch data was collected by web scraping the Wikipedia Falcon 9 launch records page.
- Using the BeautifulSoup library, the HTML tables were parsed, relevant launch information was extracted, cleaned, and converted into a Pandas DataFrame.

GitHub URL of the notebook:

[https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon 9 Data %20Collection webscraping.ipynb](https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon%209%20Data%20Collection%20webscraping.ipynb)

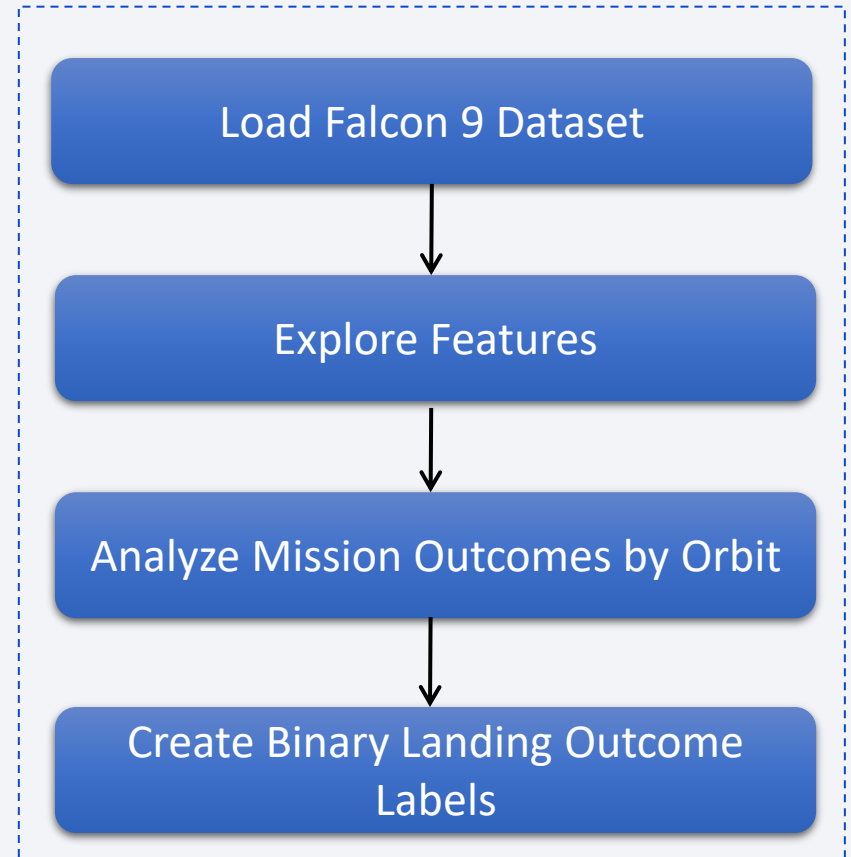


Data Wrangling

- The SpaceX launch dataset was processed to better understand the data and then prepare it for analysis accordingly.
- We examined the number of launches at each launch site, identified numerical and categorical features, analyzed the distribution of mission outcomes across different orbits.
- We created binary landing outcome labels (Class) and exported the result into csv file.

GitHub URL of the notebook:

https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/7d6f274d09cd146ff3f5e64c155f33ebe4a3df4b/Falcon_9_Data_Wrangling.ipynb



EDA with Data Visualization

Charts created:

- **Flight Number vs. Launch Site (Scatter Chart):** Compared launch frequency and success across different launch sites.
- **Payload Mass vs. Launch Site (Scatter Chart):** Analyzed payload distributions and launch outcomes for each launch site.
- **Success Rate by Orbit Type (Bar Chart):** Compared landing success rates across different orbit types.
- **Flight Number vs. Orbit Type:** Studied the relationship between launch sequence, orbit type, and landing success.
- **Payload Mass vs. Orbit Type:** Analyzed how payload mass affected landing success for different orbit types.
- **Yearly Launch Success Trend (Line Chart):** Visualized how Falcon 9 landing success improved from 2013 to 2020.

GitHub URL of the notebook: https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon_9_EDA_Visualization.ipynb

EDA with SQL

After loading the data into database, we ran the SQL queries and:

- Fetched **distinct launch sites** from the SpaceX launch dataset.
- Filtered and displayed **launch records** for launch sites beginning with "**CCA**".
- Calculated the **total payload mass** carried for **NASA (CRS)** missions.
- Computed the **average payload mass** for the **Falcon 9 v1.1** booster.
- Identified the **earliest successful ground pad landing**.
- Retrieved **booster versions** that successfully landed on a **drone ship** with payloads between **4000 kg** and **6000 kg**.
- Calculated the **number of launches for successful and failure mission outcome**.
- Identified the **booster version** used for the mission carrying the **maximum payload mass**.
- Retrieved **drone ship landing failures during 2015**, including launch site and booster version.

GitHub URL of the notebook: https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon_9_sql.ipynb

Build an Interactive Map with Folium



- **Created a Folium map** centered on the NASA Johnson Space Center as the base map.
- **Added circle markers and labeled markers** for each Falcon 9 launch site to display their locations.
- **Used a MarkerCluster** to group launch records and avoid overlapping markers.
- **Colored launch markers** using **green** for successful landings and **red** for unsuccessful landings.
- **Calculated the distance** of launch site from nearby coastline and **added distance lines and markers** to nearby coastline.

GitHub URL of the notebook: https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon_9_Interactive_Visualization.ipynb

Build a Dashboard with Plotly Dash

Dashboard Components-

- **Interactive Pie Chart:** Displays overall launch success across all sites or the success vs. failure distribution for the selected launch site, providing a quick summary of launch success rates.
- **Interactive Scatter Plot:** Visualizes the relationship between payload mass and landing success, with booster versions distinguished by different colors, making it easier to observe trends and patterns.
- **Launch Site Dropdown:** Enables users to select all launch sites or an individual launch site for analysis, allowing comparison of launch performance across different launch sites.
- **Payload Mass Range Slider:** Filters launch records based on the selected payload mass range, providing dynamic filtering of payload masses.

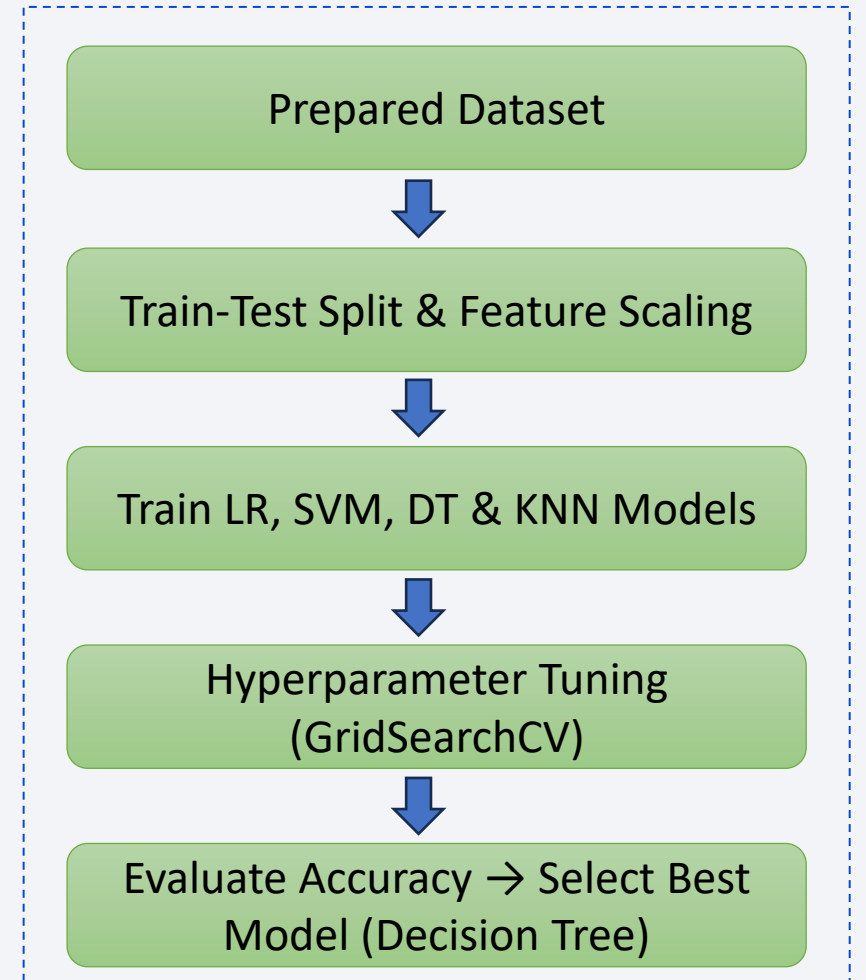
GitHub URL of the python file: [https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon 9 dash.py](https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon%209%20dash.py)

Predictive Analysis (Classification)

- Standardized the dataset and split it into training and testing sets.
- Built **Logistic Regression, SVM, Decision Tree, and KNN** classification models.
- Optimized model parameters using **GridSearchCV (10-fold cross-validation)**.
- Compared model performance using **cross-validation accuracy, test accuracy, and confusion matrices**.
- **Decision Tree** achieved the highest cross-validation accuracy and was selected as the best-performing model.

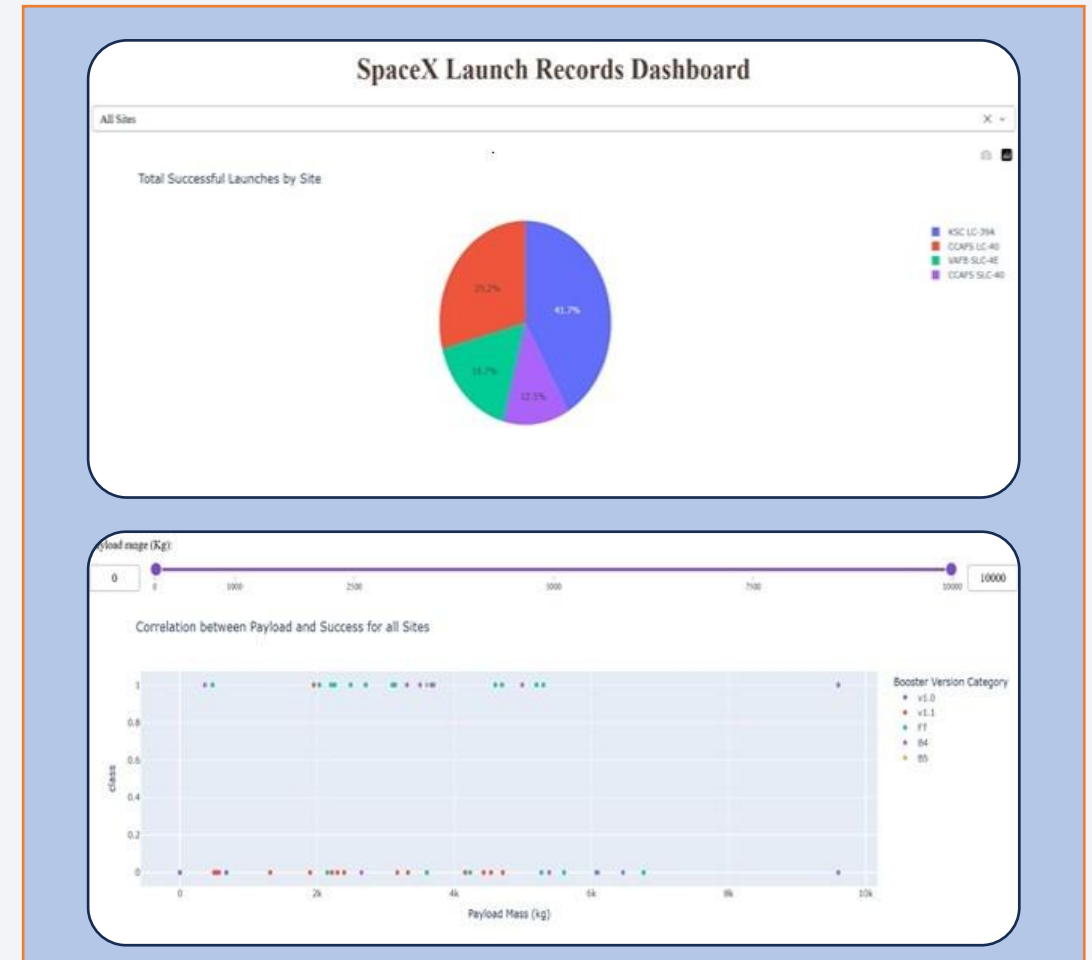
GitHub URL of the notebook:

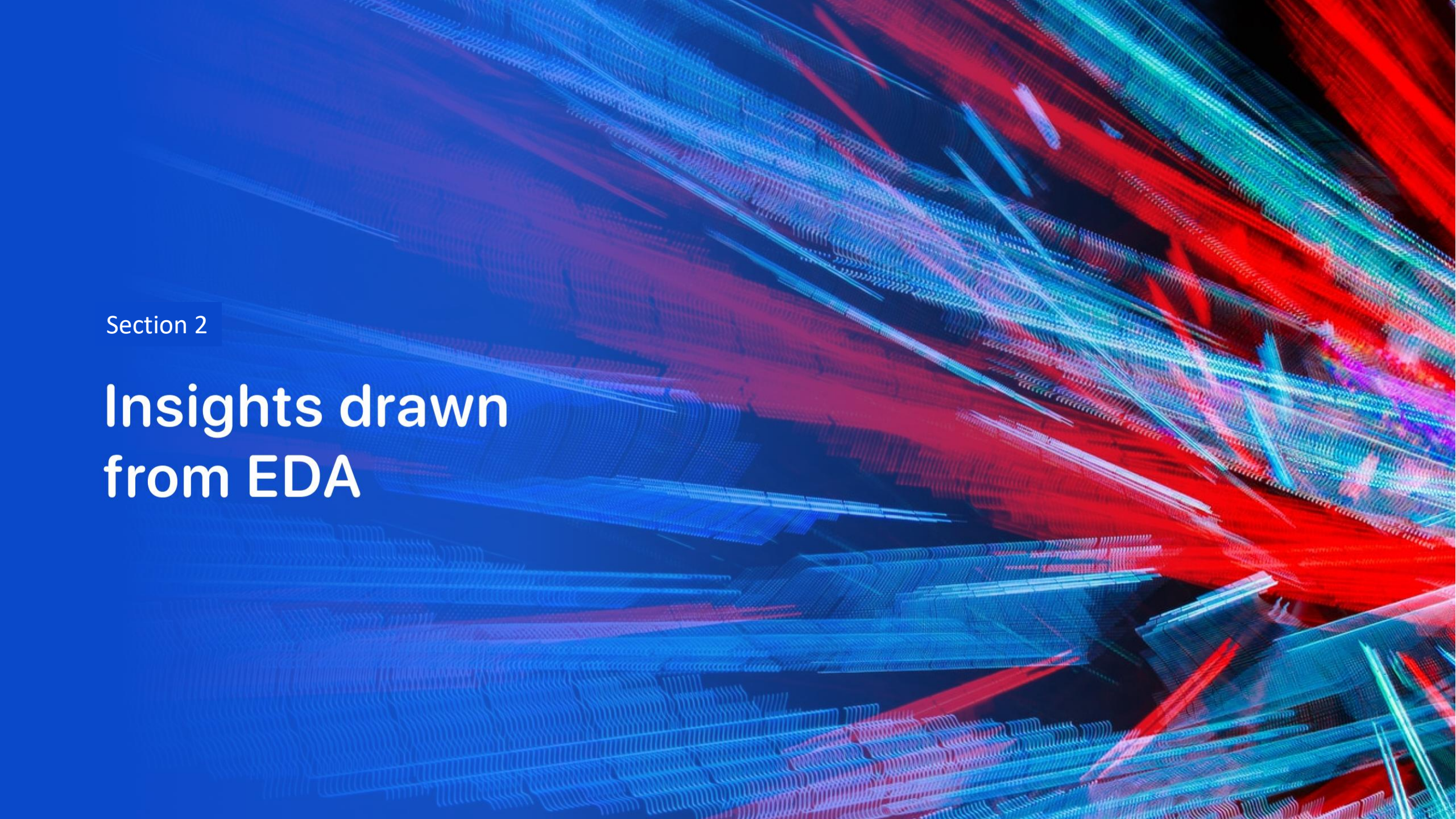
https://github.com/gravionic/IBM-Data-Science-SpaceX-Capstone-Project/blob/88203b7d93e452664911c30331960f8e1f737047/Falcon_9_ML_Analysis.ipynb



Results

- **EDA Results:** Launch success was influenced by payload mass, launch site, booster version, and orbit type.
- **Interactive Analytics:** Plotly Dash dashboard enabled dynamic exploration of launch sites, success rates, and payload trends.
- **Predictive Analysis:** Decision Tree achieved the best performance with **87.5% cross-validation accuracy** and **83.3% test accuracy**, making it the selected model.



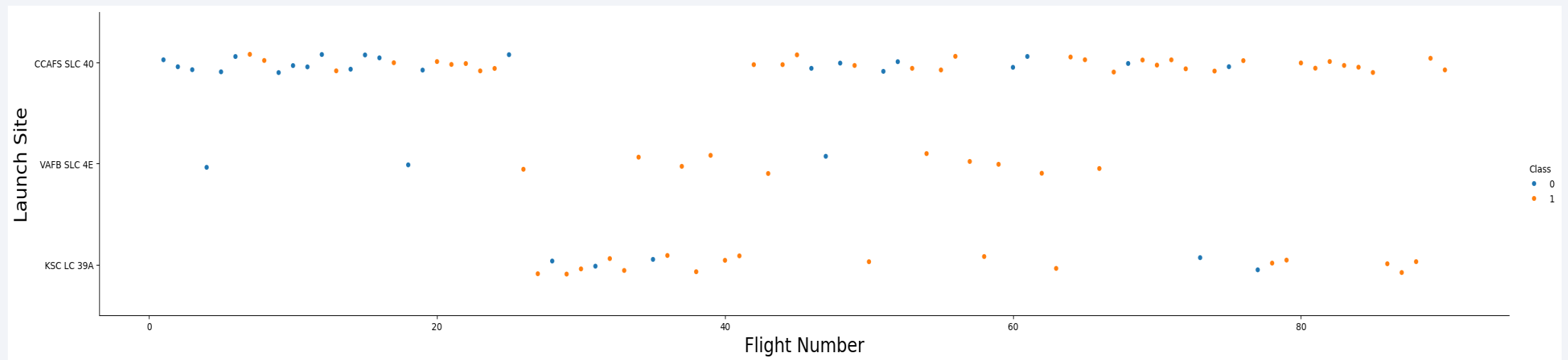
The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

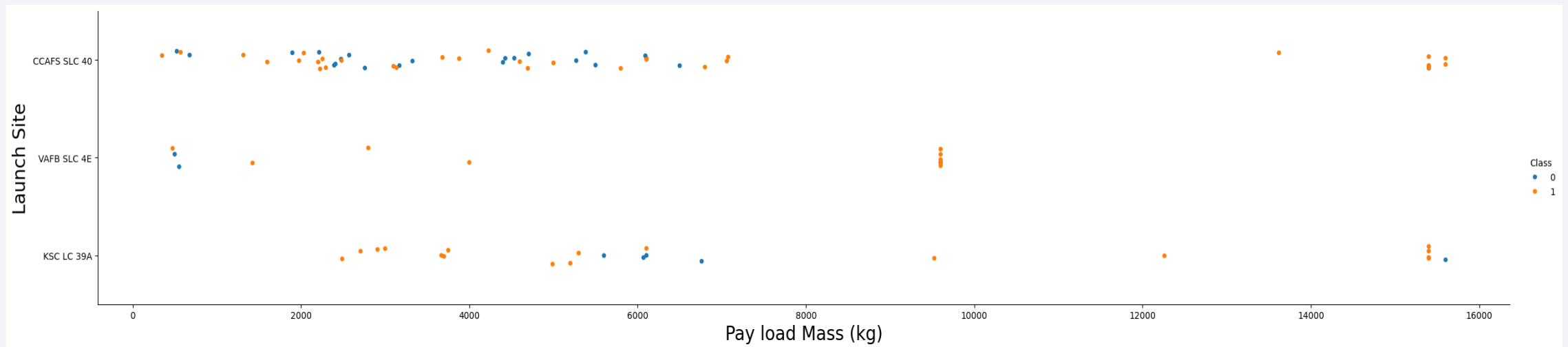
Flight Number vs. Launch Site

Here we see that as the flight number increases, the first stage is more likely to land successfully. And most successful launches were launched from CCAFS SLC 40 which also has the most number of launches ever.



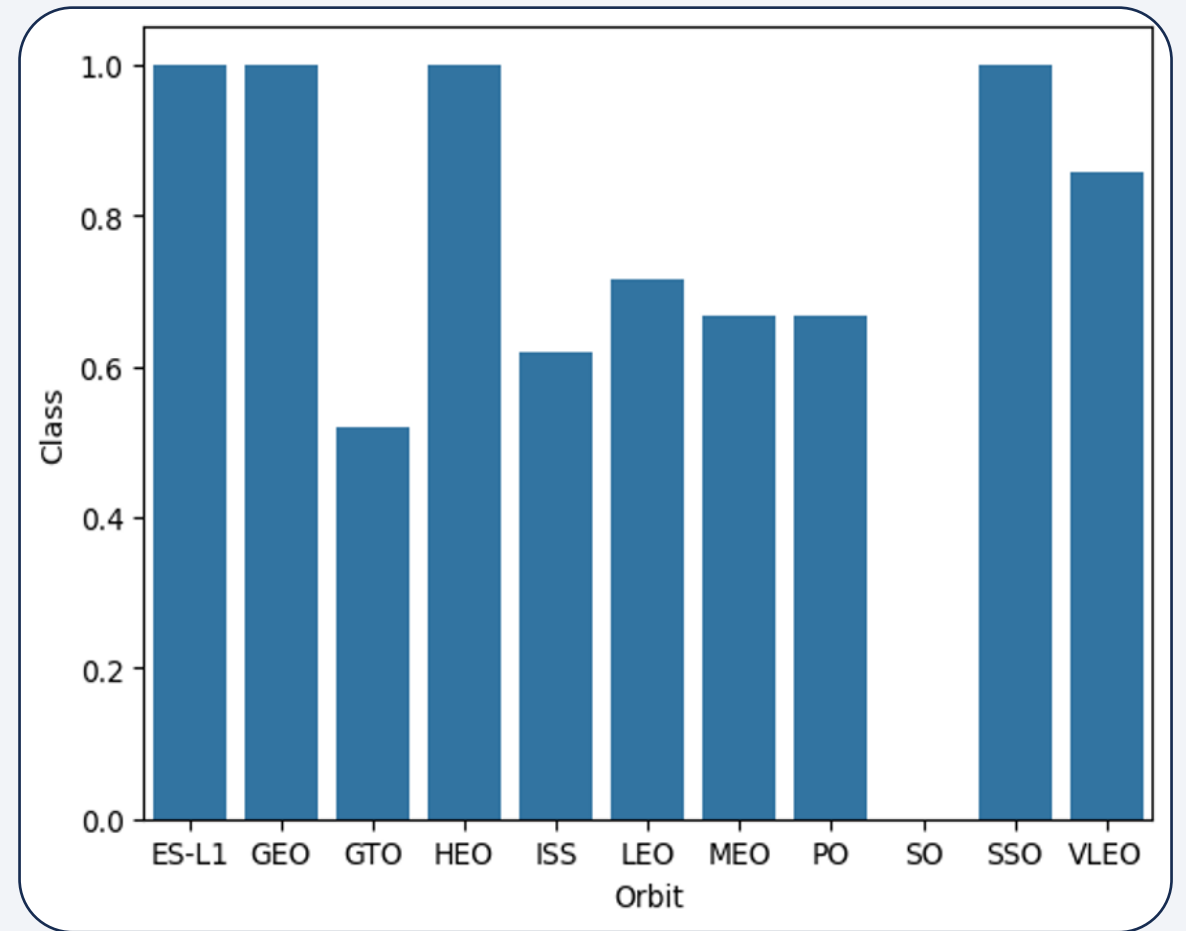
Payload vs. Launch Site

Now if we observe Payload Mass Vs. Launch Site scatter point chart we will see that for the VAFB-SLC 4E launch site there are no rockets launched for heavypayload mass(greater than 10000). And the most preferred launching sites are CCAFS SLC 40 and KSC LC 39A.



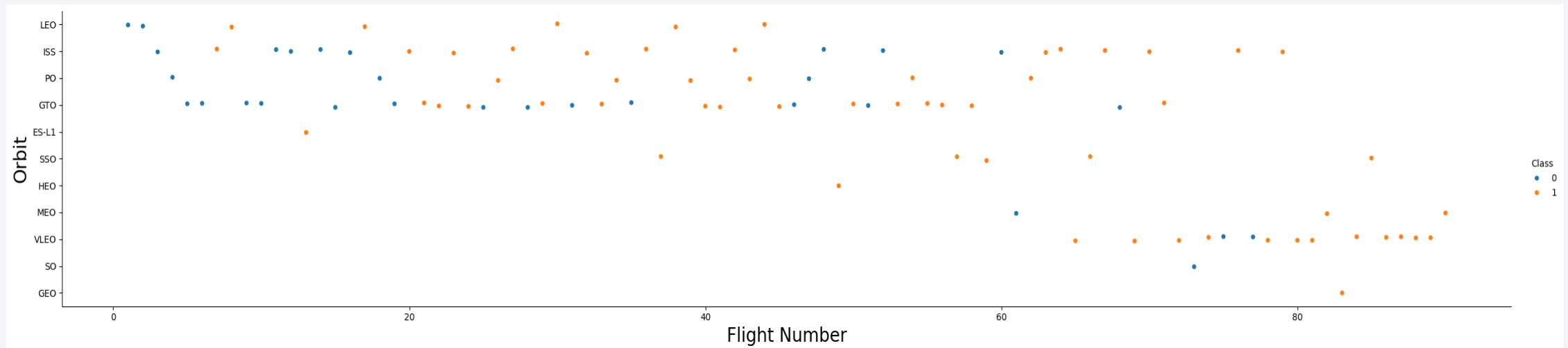
Success Rate vs. Orbit Type

Mission success differs by orbit type, with ES-L1, GEO, HEO, and SSO showing the highest success rates, while GTO and SO have the lowest among the major orbit categories.



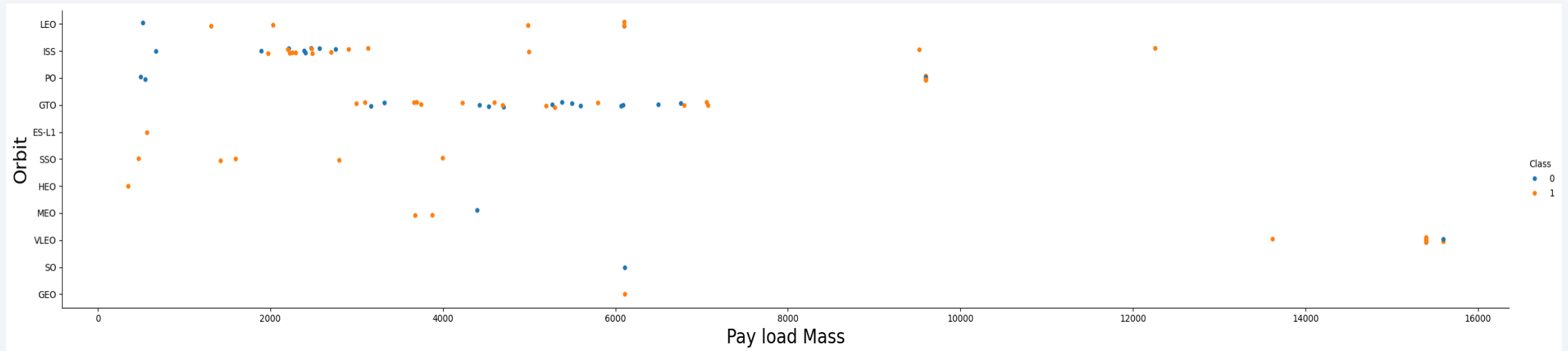
Flight Number vs. Orbit Type

As the number of flights increased over time, SpaceX expanded missions to a wider variety of orbit types. And with most number of successful flights to ISS and VLEO.



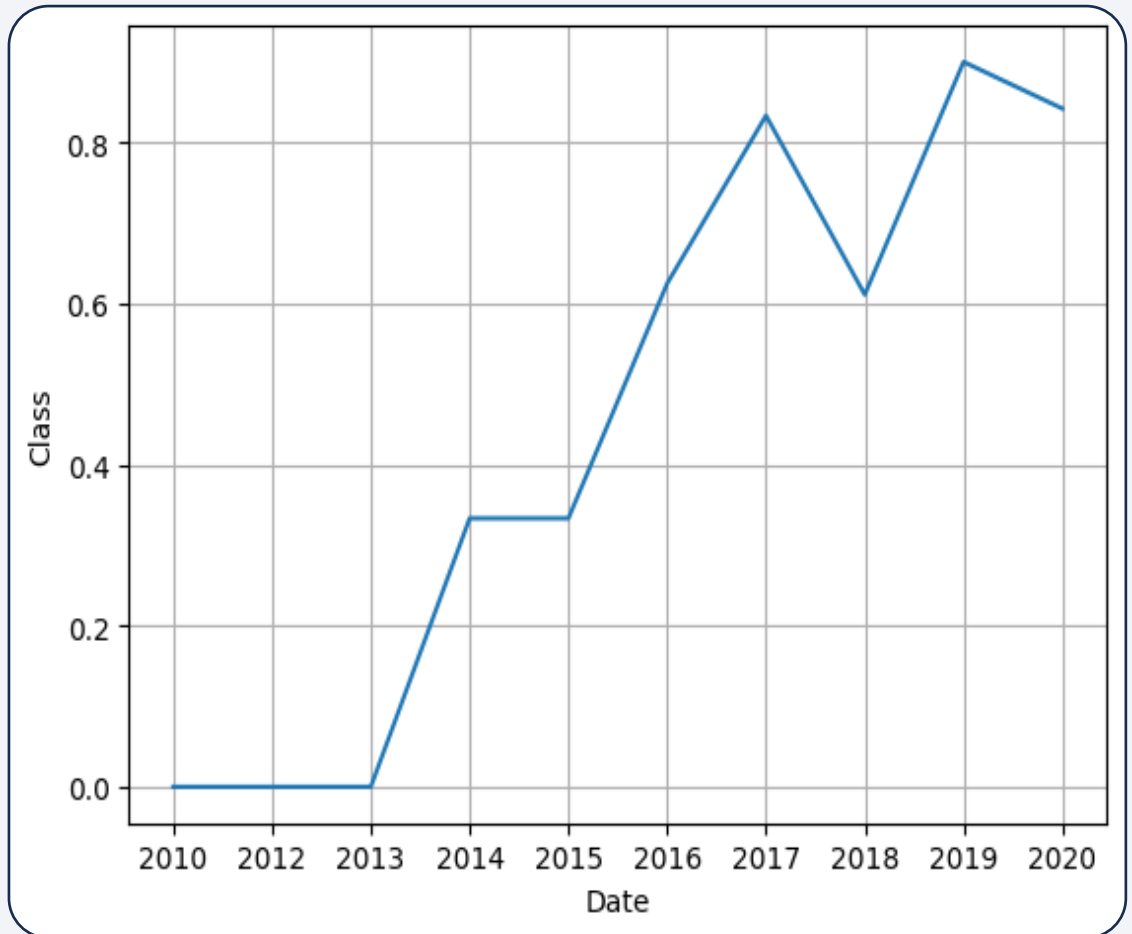
Payload vs. Orbit Type

- Different orbit types require different payload masses, with heavier payloads generally launched to specific orbits. With heavy payloads the successful landing or positive landing rate are more for LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

We can observe that the success rate since 2013 kept increasing till 2020, reflecting increased reliability of Falcon 9 missions.



All Launch Site Names

The **DISTINCT** keyword was used to return only unique values from the **LAUNCH_SITE** column, identifying the four launch sites used in the dataset.

```
%sql SELECT DISTINCT LAUNCH_SITE FROM SPACEXTABLE;
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

The LIKE 'CCA%' operator was used in the WHERE clause to filter launch sites whose names begin with "CCA" and the LIMIT 5 clause returned the first five matching records.

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Python

```
* sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

The SUM() function calculated the total payload mass carried for NASA (CRS) missions, resulting in 45,596 kg.

```
%sql SELECT sum(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Customer= 'NASA (CRS)'
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
sum(PAYLOAD_MASS_KG_)
```

```
45596
```

Average Payload Mass by F9 v1.1

The AVG() function calculated the average payload mass for the F9 v1.1 booster version, which was 2,928.4 kg.

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version= 'F9 v1.1'
```

```
* sqlite:///my\_data1.db  
Done.
```

AVG(PAYLOAD_MASS_KG_)
2928.4

First Successful Ground Landing Date

The MIN() function identified the earliest successful ground pad landing, which occurred on 2015-12-22.

```
%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome= 'Success (ground pad)'  
[80]  
... * sqlite:///my_data1.db  
Done.  
...  
MIN(Date)  
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

The query filtered successful drone ship landings with payloads between 4,000 and 6,000 kg using AND condition and the range provided and returned the matching booster versions.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome= 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;  
  
* sqlite:///my_data1.db  
Done.  
  
Booster_Version  
F9 FT B1022  
F9 FT B1026  
F9 FT B1021.2  
F9 FT B1031.2
```


Total Number of Successful and Failure Mission Outcomes

The COUNT() and GROUP BY clauses counted the number of missions for each mission outcome category.

```
%sql SELECT Mission_Outcome, COUNT(Mission_Outcome) AS Total_Successful_Missions FROM SPACEXTABLE GROUP BY Mission_Outcome;
✓ 0.0s
* sqlite:///my\_data1.db
Done.
```

Mission_Outcome	Total_Successful_Missions
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

The MAX() function identified the highest payload mass, and the matching booster versions were retrieved using a subquery.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ =(SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
✓ 0.0s

* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

The query filtered failed drone ship landings in 2015 and displayed their booster versions and launch sites.

```
%sql SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome= 'Failure (drone ship)' AND substr(Date,1,4)='2015';
```

✓ 0.0s

Python

* [sqlite:///my_data1.db](#)

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

The query used COUNT(), GROUP BY, and ORDER BY DESC to rank landing outcomes by frequency between 2010 and 2017.

```
%sql SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC;
✓ 0.0s
* sqlite:///my_data1.db
Done.
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Global Distribution of SpaceX Launch Sites

The map displays all SpaceX launch sites, showing that they are concentrated in the United States near coastal regions.



Launch Outcomes at SpaceX Launch Sites

Green markers indicate successful launches, while red markers represent failed launches at each launch site.

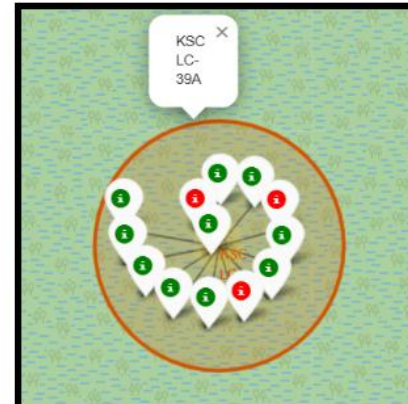
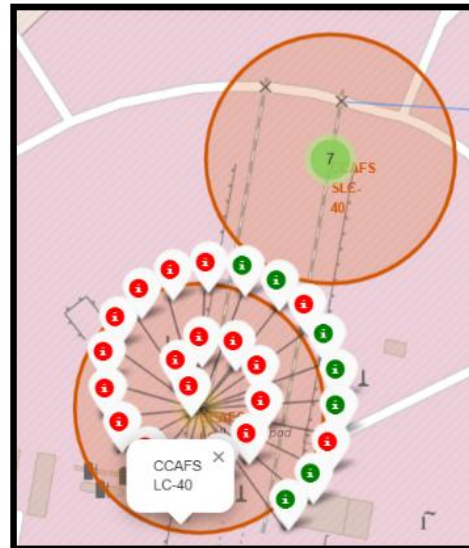
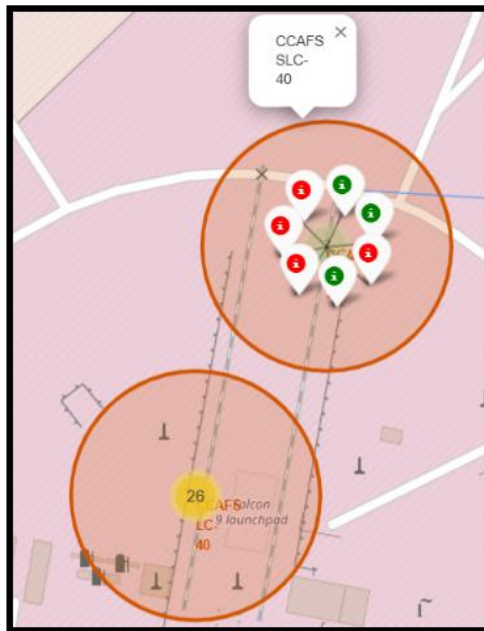


Fig: Launch Sites in Florida

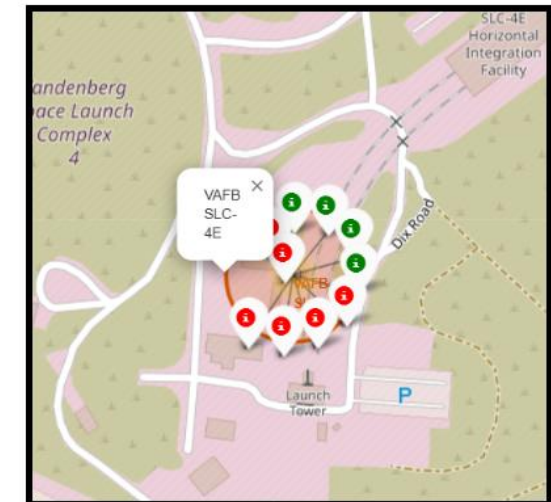


Fig: Launch Site in California

Launch Site Proximity Analysis

The map measures the distance from launch sites to nearby highways, railways, coastlines, and cities, highlighting their strategic locations for accessibility and safety.



Fig: Distance to nearest Highway and the Coastline



Fig: Distance to nearest Railwayline

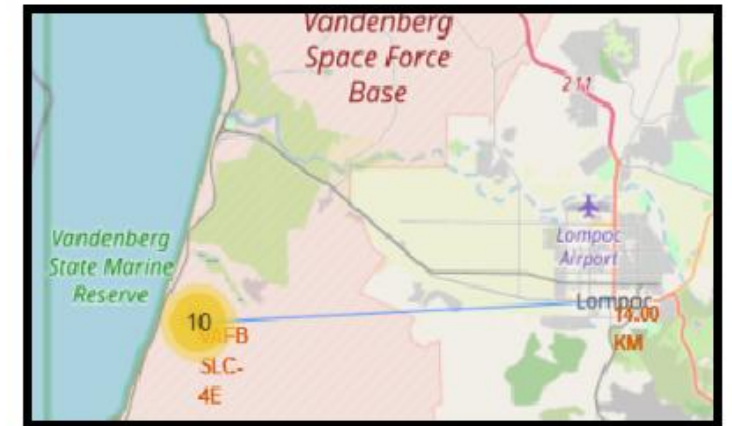
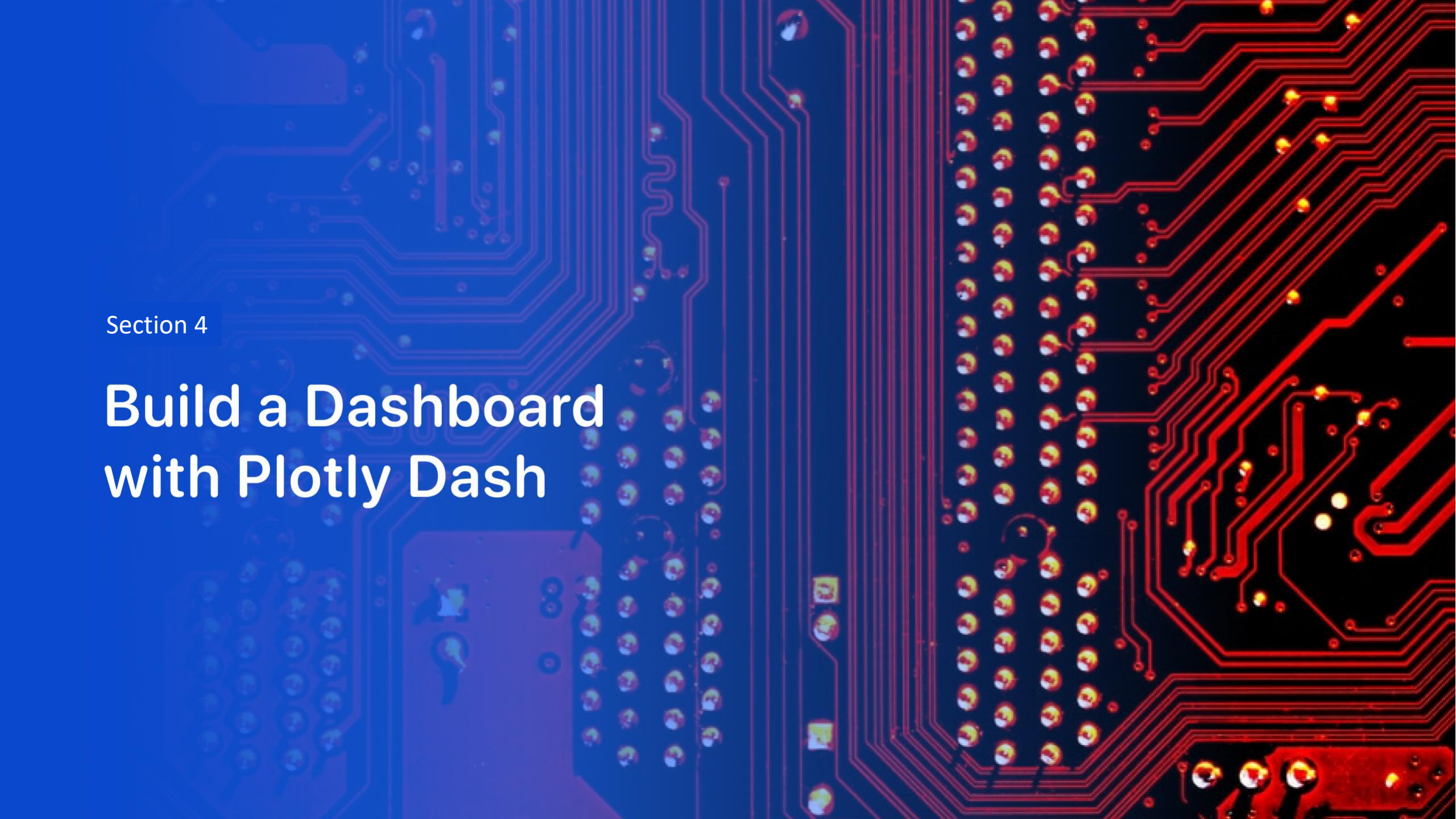


Fig: Distance to nearest City

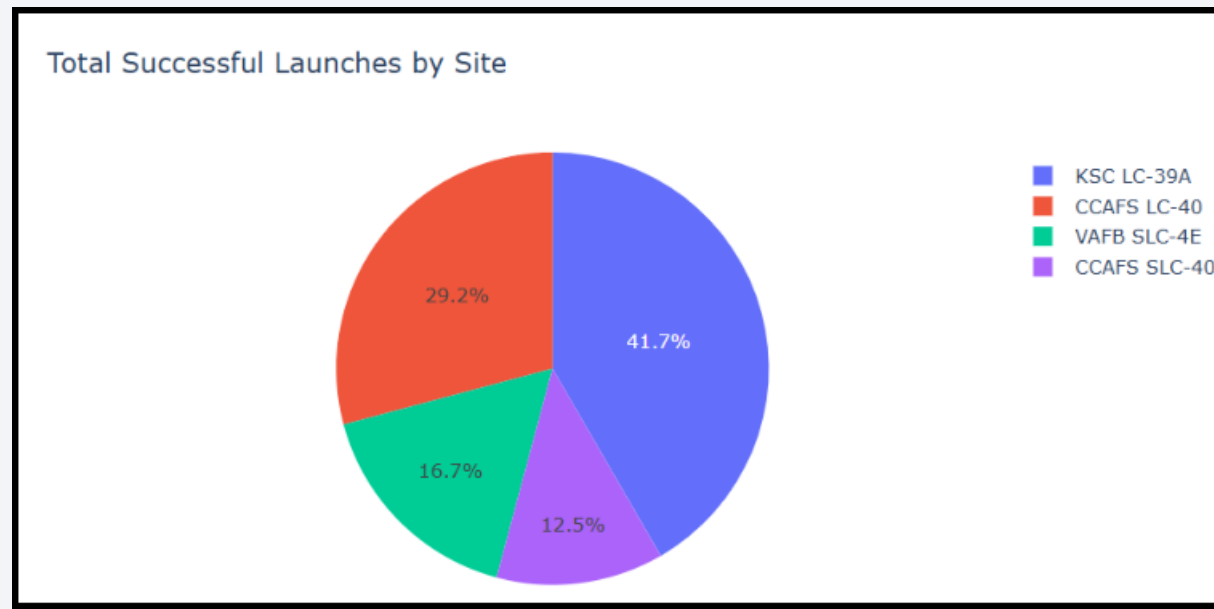


Section 4

Build a Dashboard with Plotly Dash

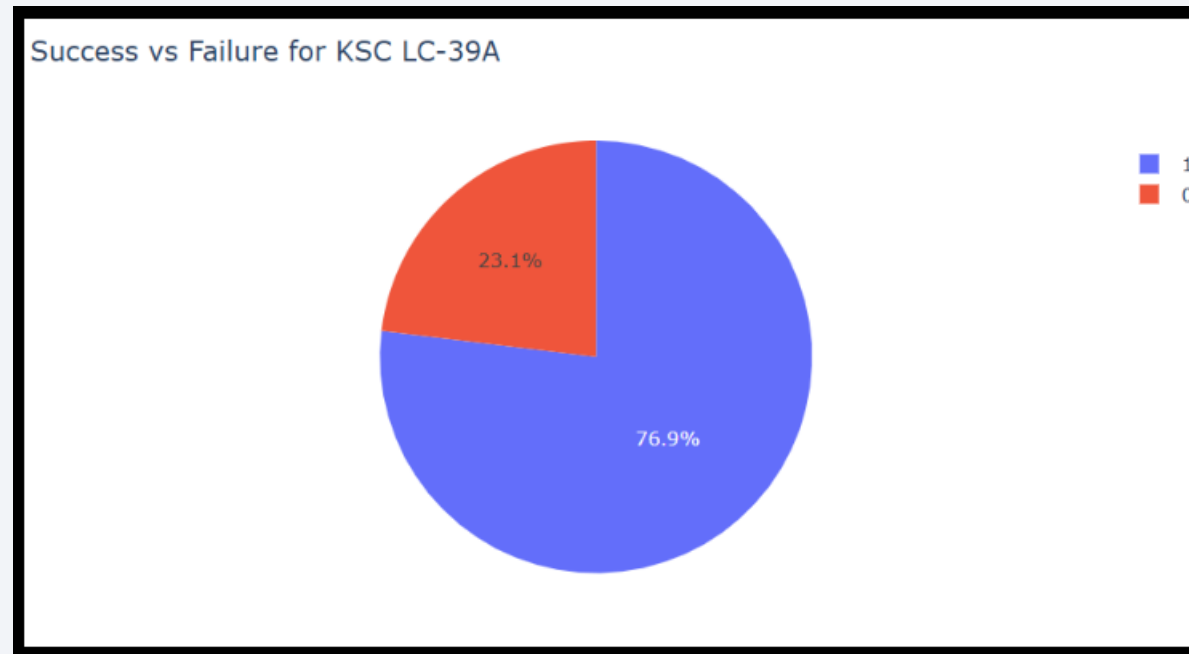
Launch Success Count for all Sites

The pie chart compares the total number of successful launches across all SpaceX launch sites. KSC LC-39A has the largest share of successful launches (41.7%), then comes CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), and CCAFS SLC-40 (12.5%).



Success vs. Failure Ratio at the Highest-Performing Launch Site

The pie chart shows the launch outcomes for KSC LC-39A, the launch site with the highest success ratio. Approximately 76.9% of launches were successful, while 23.1% resulted in failure, showing the site's strong reliability.



Payload Mass vs. Launch Success Analysis

The lower payload range (0-5000 kg) has a higher success rate, while the higher payload range (5000-10000 kg) includes relatively more failed launches. Newer booster versions, such as FT and B4, also exhibit improved success rates at higher payload masses.



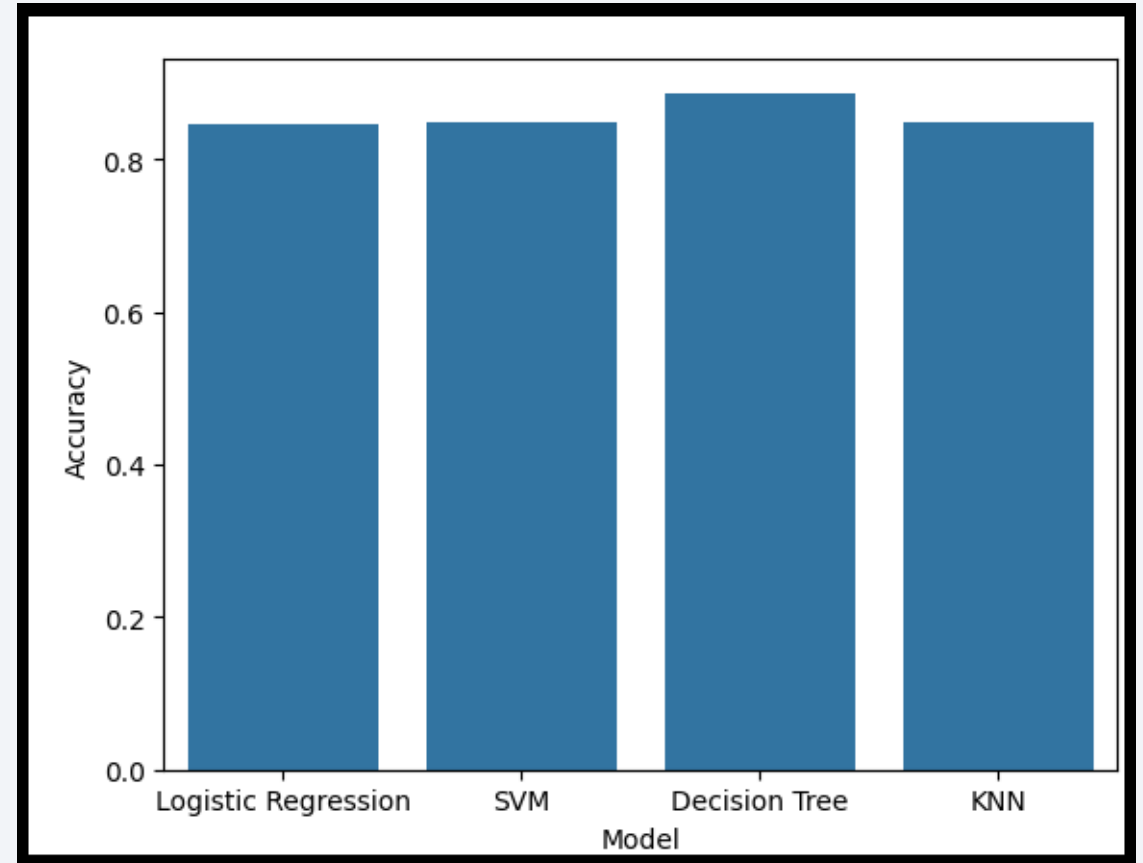


Section 5

Predictive Analysis (Classification)

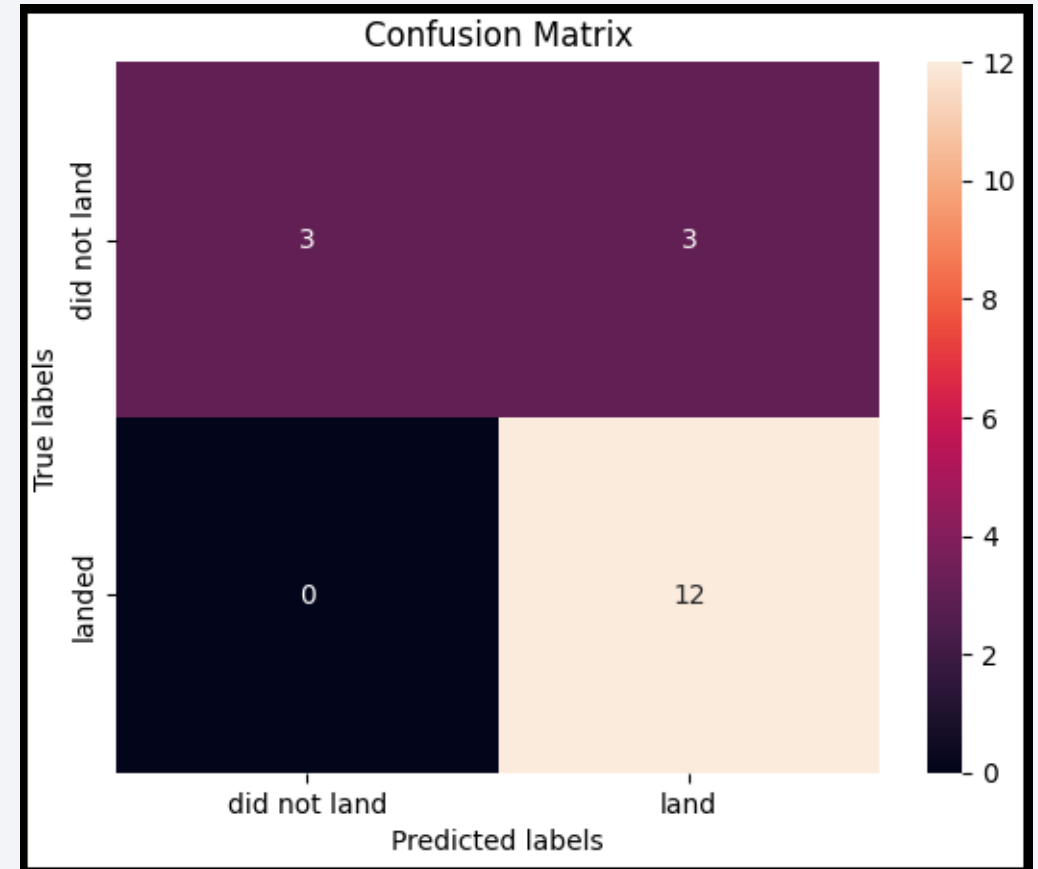
Classification Accuracy

Four classification models were evaluated that are Logistic Regression, SVM, Decision Tree, and KNN. Among them, the **Decision Tree** model achieved the highest cross-validation accuracy of **87.5%**, making it the best-performing model for predicting Falcon 9 landing outcomes.



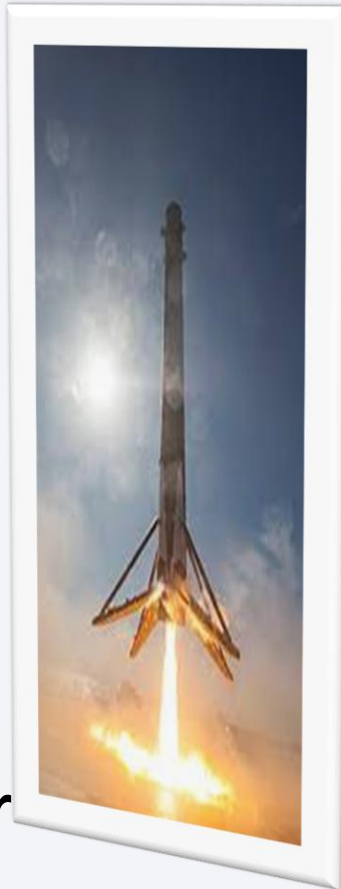
Confusion Matrix

The confusion matrix shows that the Decision Tree model correctly classified **12 successful landings** and **3 failed landings**, with **3 false positives** and **no false negatives**, indicating good overall predictive performance, especially in the cases of successful launches.



Conclusions

- Launch site **KSC LC-39A** recorded the highest number of successful launches.
- Missions to orbits **ES-L1, GEO, HEO, and SSO** achieved the highest landing success rates.
- **Lower payload masses** generally showed higher landing success than heavier payloads.
- **Decision Tree** was the best-performing classifier with **87.5%** cross-validation accuracy.
- The analysis shows that **launch site, orbit, payload mass, and booster version** are key factors affecting Falcon 9 first-stage landing success.



Thank you!

